A Comparative Study of Supervised and Semi-Supervised Retraining for Drift-Aware Malware Detection

Tahsin Kabir
Department of CSE, BUET
2005064@ugrad.cse.buet.ac.bd

Shanto Mondol
Department of CSE, BUET
2005073@ugrad.cse.buet.ac.bd

Md. Mostofa Akbar
Department of CSE, BUET
mostofa@cse.buet.ac.bd

***Abstract*—Mobile malware keeps evolving, constantly adopting new behaviors and blending in with benign apps. Over time, this shifts the underlying feature distribution and creates concept drift, which steadily erodes the performance of classifiers. Periodic retraining helps, but collecting and labeling fresh samples at scale is expensive and slow in the real world. In this work, we carried out a comparative study of supervised and semi-supervised retraining strategies to handle drift in Android malware detection. Using a large temporally ordered dataset, we explored whether selectively retraining on drifted samples could keep detection performance high while cutting down the cost of manual labeling. To find the most informative samples, we tested several drift and anomaly detection methods. We also examined pseudo-labeling as a lighter semi-supervised alternative and looked closely at the trade-off between accuracy and labeling effort. Our experiments showed that drift-aware incremental retraining consistently beats static models when malware distributions change over time.**

***Index Terms*—concept drift, malware evolution, drift-aware retraining**

I. Introduction

Machine learning has become a go-to approach for Android malware detection because it can automatically pick up patterns from huge collections of apps. However, models trained on older data gradually lose effectiveness when faced with apps collected months or years later due to concept drift. The obvious fix is to retrain the model regularly. But this quickly becomes expensive, hard to scale and retraining on every new sample is wasteful. A smarter strategy is to carefully pick the most informative samples to keep labeling costs under control.

In this work, we explore drift-aware retraining strategies and test several drift and anomaly detection techniques to select useful samples for retraining. To structure our study, we focus on three key research questions:

RQ1: How effective are different drift and anomaly detection methods at identifying informative samples for retraining as malware distributions evolve?

RQ2: Can semi-supervised retraining with pseudo-labels deliver performance close to fully supervised retraining while significantly reducing the labeling burden?

RQ3: How much does drift-aware sample selection improve the balance between detection performance and labeling cost compared to traditional retraining approaches?

II. Related Works

Existing approaches can be broadly categorized into drift detection, classifier adaptation, and drift anticipation. Among drift detectors, CADE [1] identifies drifted samples using constrastive autoencoder & class-centric distance thresholds. DREAM [2] jointly optimizes drift detection and classification to better capture intra-family evolution. Guided Retraining [3] prioritizes difficult samples during retraining, whereas MORPH [4] and ADAPT [5] leverage pseudo-labels to reduce labeling costs. Other approaches include test-time adaptation in MADCAT [6] and reinforcement learning-based sample selection in DRMD [7]. Drift anticipation has also been explored using generative models that synthesize future malware behaviors.

Unlike prior work, we compare CADE against six classical and deep anomaly detectors, and analyze the effectiveness of drift-aware sample selection relative to full retraining. To the best of our knowledge, existing studies have not systematically compared these dimensions together using a single long-term temporal evaluation protocol.

III. Methodology

Let (T_0) denote the initial training period (year 2012) and (T_m) denote the malware samples arriving in month (m) from January 2013 to November 2018. Using the APIGraph dataset, we first train a baseline XGBoost classifier and seven drift/anomaly detectors on (T_0): CADE, DeepSAD, Isolation Forest (IF), One-Class SVM (OC-SVM), Local Outlier Factor (LOF), Mahalanobis Distance, and PCA Reconstruction Error.

For each incoming month (T_m), samples identified as drifted by the detectors are selected for model adaptation. We evaluate both supervised and semi-supervised retraining strategies. In the supervised setting, selected samples are retrained using their ground-truth labels. In the semi-supervised setting, pseudo-labels are assigned to drifted samples before retraining. For CADE, pseudo-labels are generated using the nearest malware-family centroid in the learned latent space.

Retraining is performed incrementally, where the selected samples from (T_m) are incorporated into the training pool and the XGBoost classifier is updated before evaluating on (T_{m+1}).

TABLE I

PERFORMANCE GAIN AND LABELING COST RELATIVE TO THE STATIC BASELINE FOR THE NOVEMBER 2015 EVALUATION WINDOW. SAMPLE RATIO DENOTES THE FRACTION OF ADDITIONAL SAMPLES USED BY EACH METHOD COMPARED TO FULL RETRAINING.

Method	**Additional Samples**	**Sample Ratio (%)**	**Accuracy**	**Acc. Increase (%)**	**F1 Macro**	**F1 Increase (%)**
Baseline	0	–	0.918	–	0.636	–
Full Retraining	154,560	100.00	0.996	7.86	0.990	35.42
CADE-True	15,617	10.10	0.979	6.09	0.935	29.95
Mahalanobis	64,404	41.67	0.990	7.20	0.970	33.47
DeepSAD	41,648	26.95	0.989	7.16	0.969	33.34
IF	22,363	14.47	0.965	4.68	0.884	24.80
OC-SVM	25,706	16.63	0.974	5.63	0.919	28.31
LOF	46,662	30.19	0.976	5.79	0.924	28.88
PCA	17,433	11.28	0.924	0.64	0.677	4.16
CADE-Pseudo	15,617	10.10	0.916	-0.14	0.614	-2.14

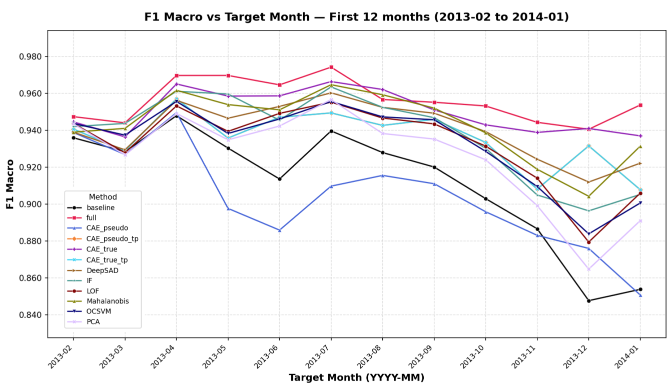

Fig. 1. Macro-F1 over the first 12 months.

IV. EXPERIMENTAL ANALYSIS

Figure 1 illustrates the F1-macro performance during the first twelve months following the initial training period. The static XGBoost baseline exhibits a gradual decline in performance, with F1 decreasing from approximately 0.94 to below 0.86 within one year. As expected, full retraining consistently achieves the highest performance, however, several drift-aware retraining strategies achieve comparable results while requiring substantially fewer samples. In particular, CADE with true labels closely follows the performance of full retraining throughout the evaluation period. Mahalanobis distance and DeepSAD also demonstrate strong adaptation capability.

Table I further quantifies the trade-off between performance and retraining cost. Full retraining achieves the highest improvement over the static baseline, increasing accuracy and F1-macro by 7.86% and 35.42%, respectively. Remarkably, CADE-True achieves a 6.09% accuracy improvement and a 29.95% F1 improvement while using only 10.1% of the samples required by full retraining. This result suggests that carefully selected drifted samples can preserve most of the adaptation benefits of exhaustive retraining while significantly reducing labeling and computational costs. Mahalanobis and DeepSAD provide competitive performance but require considerably larger retraining sets.

V. DISCUSSION

The experiments reveal that the effectiveness of retraining depends more on sample quality than sample quantity. Although full retraining provides the highest performance, it incurs substantial labeling and computational overhead. In contrast, drift-aware sample selection identifies a compact set of informative samples that capture the evolving malware distribution. Among the evaluated approaches, CADE achieves the best balance between performance and cost while using approximately one-tenth of the additional samples required by full retraining. These findings suggest that selective retraining is a practical strategy for maintaining long-term malware detection systems under concept drift. The weaker performance of pseudo-label-based retraining indicates that label noise remains a significant challenge in semi-supervised adaptation. Future work will investigate more robust pseudo-labeling techniques.

ACKNOWLEDGMENT

The authors thank their supervisor and collaborators at the Department of CSE, BUET, for guidance.

REFERENCES

[1] L. Yang, W. Guo, Q. Hao, A. Ciptadi, A. Ahmadzadeh, X. Xing, and G. Wang, "CADE: Detecting and explaining concept drift samples for security applications," in *Proc. 30th USENIX Security Symposium (USENIX Security '21)*, Aug. 2021, pp. 2327–2344.

[2] Y. He, J. Lei, Z. Qin, K. Ren, and C. Chen, "Combating concept drift with explanatory detection and adaptation for Android malware classification," in *Proc. ACM SIGSAC Conf. Computer and Communications Security (CCS '25)*, Taipei, Taiwan, Oct. 2025.

[3] N. Daoudi, K. Allix, T. F. Bissyandé, and J. Klein, "Guided retraining to enhance the detection of difficult Android malware," in *Proc. 31st ACM SIGSOFT (ISSTA '22)*, 2022, pp. 1131–1143.

[4] M. T. Alam, R. Fieblinger, A. Mahara, and N. Rastogi, "Poster: MORPH – Towards automated concept drift adaptation for malware detection," in *Proc. Network and Distributed System Security Symposium (NDSS '24) Posters*, 2024.

[5] M. T. Alam, A. Piplai, and N. Rastogi, "ADAPT: A pseudo-labeling approach to combat concept drift in malware detection," arXiv preprint arXiv:2507.08597, 2025.

[6] E. Roh, Y. Kaya, C. Kruegel, G. Vigna, and S. Hong, "MADCAT: Combating malware detection under concept drift with test-time adaptation," arXiv preprint arXiv:2505.18734, 2025.

[7] S. McFadden, M. Foley, M. D'Onghia, C. Hicks, V. Mavroudis, N. Paoletti, and F. Pierazzi, "DRMD: Deep reinforcement learning for malware detection under concept drift," in *Proc. 40th AAAI Conf. on Artificial Intelligence (AAAI '26)*, 2026.